Math 526, F08 **Name:** Answer

Quiz 5 **SSN: xxx-xx-**

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (8 points) Let $\mathbf{A} = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ -2 & -6 & 0 & 0 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

(a) Reduce $[\mathbf{A}\ \ \mathbf{b}]$ to $[\mathbf{R}\ \ \mathbf{d}]$ to find the condition on b_1, b_2, b_3 such that $\mathbf{Ax} = \mathbf{b}$ is solvable.

Solution: $[A\ \ b] = \left[\begin{array}{cccc|c} 1 & 3 & 1 & 2 & b_1 \\ 2 & 6 & 4 & 8 & b_2 \\ -2 & -6 & 0 & 0 & b_3 \end{array}\right]$

$\to \left[\begin{array}{cccc|c} 1 & 3 & 1 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2-2b_1 \\ 0 & 0 & 2 & 4 & b_3+2b_1 \end{array}\right]$

$\to \left[\begin{array}{cccc|c} 1 & 3 & 1 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2-2b_1 \\ 0 & 0 & 0 & 0 & b_3+4b_1-b_2 \end{array}\right]$

$\to \left[\begin{array}{cccc|c} 1 & 3 & 0 & 0 & 2b_1-\frac{1}{2}b_2 \\ 0 & 0 & 2 & 4 & b_2-2b_1 \\ 0 & 0 & 0 & 0 & b_3+4b_1-b_2 \end{array}\right]$

$\to \left[\begin{array}{cccc|c} 1 & 3 & 0 & 0 & 2b_1-\frac{1}{2}b_2 \\ 0 & 0 & 1 & 2 & \frac{b_2}{2}-b_1 \\ 0 & 0 & 0 & 0 & b_3+4b_1-b_2 \end{array}\right] = [R\ \ d]$

(b) Find the complete solution to $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$.

Solution: $[R\ \ d] = \left[\begin{array}{cccc|c} 1 & 3 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 2 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$

Particular solution: $x_2 = x_4 = 0$, $Rx =$

$$\Rightarrow x_1 = \frac{1}{2}, \quad x_3 = \frac{1}{2}$$

$$\Rightarrow x_p = \left(\frac{1}{2}, 0, \frac{1}{2}, 0\right)$$

special solution: Solve $Rx = 0$

i) $x_2 = 1$, $x_4 = 0 \Rightarrow x_1 = -3, x_3 = 0$

$$\Rightarrow s_1 = (-3, 1, 0, 0)$$

ii) $x_2 = 0$, $x_4 = 1 \Rightarrow x_1 = 0, x_3 = -2$

$$\Rightarrow s_2 = (0, 0, -2, 1)$$

Complete solution to $Ax = b$ is

$$x = x_p + x_n$$

$$= \begin{bmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$